Preconception Nutrition Beyond Folate Is Not a Second NTD Trial

Reviews of preconception nutritional risks find a mixed, nutrient-specific evidence base; folate's High/stable NTD identification does not generalize to other preconception nutrients or to later IQ.

W-TO-MODERATE-BEYOND-FOLATE; LOW-IQ-CHECKLIST | STATUS: DRAFT | DOMAIN: PRECONCEPTION, NUTRITION | REPLICATION

WHAT WE KNOW (Narrowest Supported Conclusion)

Dean and colleagues (2014) reviewed nutritional risks and interventions in the preconception period. Evidence is nutrient-specific and often thin. Periconceptional folic acid for neural-tube defects is the strong identification — already prenatal-folate-ntd.md, status: stable for that narrow claim. Other nutrients (iron, calcium, iodine in replete populations, multivitamins as a package) do not inherit that grade.

EVIDENCE AND CAUSAL ASSESSMENT

Narrative/systematic review of mixed trials and observational studies, not a new RCT. Adjacent: prenatal-folate-ntd.md remains the NTD card; do not re-estimate it. prenatal-iodine-thyroid.md is pregnancy deficiency, not a preconception IQ program. Paternal-age mutation rate (paternal-age-de-novo-mutations.md) is a different preconception object. Sperm-epigenetic lifestyle stories remain excluded as a standalone card. Sleep and attachment are not restated.

EFFECT SIZE AND PERSISTENCE

Folate's NTD risk reduction is large and already graded High/stable. Other reviewed interventions have smaller, inconsistent, or baseline-dependent effects. Persistence to childhood cognition is generally untested or confounded.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (nutritional epidemiology and trials, not an observational population phenotype).

Baseline deficiency determines marginal benefit. Trials in deficient populations do not license supplements in replete ones. Observational diet–IQ links mix education, income, and health care.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Preconception biology is not parental control. A missed supplement is not a script for later intelligence. Folate's success is not a license for every nutrient. Animal epigenetic marking is not a childhood.

WHAT ACTION IS REASONABLY JUSTIFIED

Keep folate's NTD identification. Do not sell a general preconception brain program. Nutrient claims beyond folate need their own designs.

VERIFIED SOURCES & DOI IDENTIFIERS

- Dean et al. (2014), Reproductive Health